Dear Reader: I Cordially Invite You to Uplevel "Behavior Analysis" as Historically Known.

Yours Truly, Artificial Intelligence

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"There is no sin punished more implacably by nature than the sin of resistance to change."

Anne Morrow Lindbergh

Change is upon us, and arguably has been for some time. It is difficult to understate the impact and increasing role that data-driven, algorithmic, and artificial intelligence (AI) based technologies have on modern societies and human behavior. Smartphones constantly track our location and movement. Web browsers track every site we visit. Websites track our swipes, scrolls, clicks, eye movements, and time spent. Wearables track every heartbeat, sweat bead, and when we go to and get out of bed. Sensors collect data on ambient lighting and temperature, adjusting our digital screens and home temperature to perfectly match our preferences. All the while, Siri, Alexa, or Cortana patiently "listen" to our every word and the soundscape around us.

Across this data landscape of our lives, algorithms quietly hum in the background. Each dutifully crunches behavioral data, serving up recommendations, nudges, reminders, productivity enhancers, attention grabbers, and offering—for a price—immediate access to social interactions, tangible items, non-socially mediated idiosyncratic experiences, and escape from, or avoidance of, the things we find less preferred. All the while, an AI companion tai-

lored to our likes and ways of thinking will chat with us about anything and everything we'd like; offer advice for anything that worries us (for better or worse); teach us about things that may have previously been out of our grasp; do our holiday and grocery shopping; and complete any text, image, or video-based task we ask it to without reverence for accuracy, truth, or environmental impact. After all, these systems are designed to please.

Welcome to 2026.

As an evolutionary science, behavior analysis draws on the concepts of selective pressure, adaptation, survival, and extinction. Yet, as the world around behavior analysis changes, is behavior analysis changing with it? The research questions and methods used in our journals look like those of 20-30 years ago. Job prospects outside academia for those without PhDs are centered on the big three: autism and developmental disabilities, education, and organizational behavior management (OBM). Leaning into the topic sentence, broad, selective pressures in society writ large are driving tremendous change in the collection and analysis of data on human behavior and preferences across all industries; creating new jobs, new job-related skills, and new tools and technologies; generating new research questions; and creating new opportunities. As day-to-day and job-related human behavior change radically, where are the representatives of the science of behavior found?

To be fair, it is good that sciences change slowly by requiring either overwhelming evidence or the passage of a generation in power. Jumping on bandwagons and newfangled ideas too quickly can distract from building a systematic, cohesive, and trustworthy foundation that future scientists can build on. As Sir Karl Popper notes, scientific change is mostly gradual, realistically cumulative, and driven primarily by the fit of gathered evidence to scientists' and practitioners' experience with the world and problem-solving effectiveness.

Sometimes, sciences change rapidly due to a punctuating event. John Grim reminds us that one way this can occur is by accumulating evidence that cannot be accounted for by existing paradigms, leading to a paradigm shift and rapid change. Alternatively, as Thomas Kuhn famously claimed, paradigm shifts occur due to near-singular events, such as technological developments. For example, Robert Hooke and Anton Van Leeuwenhoek showed how the microscope enabled humans to see universes of organisms entirely

new to us and led to the development of cell theory. Gene sequencing gave humans the ability to physically see and gather data on what had previously been primarily logico-mathematical descriptions of heredity, subsequently changing how we see *genus-species-subspecies* relations. And computers gave humans the ability to automate symbolic and numerical operations at scale, simulate complex systems, and represent data and algorithms in a unified, programmable form. This, in turn, changed our understanding of mathematical and computational relations fundamental to nearly all sciences as described well by Alan Turing and John Von Neumann.

We have our own technological punctuation in behavior science. The development of the operant chamber allowed humans to isolate and observe the unique contributions of response-consequence relations as distinct (though not entirely separable from) stimulus-stimulus and stimulus-consequence relations. Here, advances in technology led to near-immediate changes in what scientists studied, how they studied their phenomena of interest, and how the next generation of behavior scientists and practitioners were educated and trained.

AI is having a similar disruptive effect in many areas of science. Deep-learning systems and computer vision in medical imaging now prescreen scans and draft preliminary reads, shifting radiologists' work to analyzing edge cases, integrating different data types, and overseeing models. Generative and large language models (LLMs) in drug discovery propose novel compounds and trial designs, shifting pharmacologists' and chemists' work from being manual searchers to algorithm supervisors and design critics. In logistics and manufacturing, computer vision and reinforcement learning guide robots to reconfigure warehouse flows, creating new roles in human-robot coordination and data-centric operations while reducing the need for traditional line work. And in education, adaptive learning platforms and AI tutors have extended B.F. Skinner's teaching machines via personalized pacing and content at scale, shifting educators' work toward data interpretation, ethical oversight, and the curation of AI-augmented learning environments. Behavior analysis is similarly being impacted, whether we recognize it or not.

Changing *What* We Study

Since the beginning of the scientific revolution, the primary driving assumption guiding research design has been reductionism by isolating phenomena in a laboratory. In physics, Galileo stripped motion of all but gravity using inclined planes, and Millikan isolated a single charged droplet to measure the elementary charge. In chemistry, Lavoisier confined reactants to closed vessels to reveal mass conservation, and Wöhler synthesized urea to show organic molecules could be isolated from "vital" contexts. And in biology, Mendel controlled hereditary traits through tightly bred pea lines, and Hodgkins and Huxley isolated a giant squid axon to derive the ionic basis of action potentials.

However, something changed during the 20th century. Despite the many advances made in our understanding of the world around us, simple failures started cropping up everywhere. In physics, reductionist models struggled with turbulent flow and the emergent behavior of complex systems. In chemistry, isolated reaction studies could not predict catalytic behavior in real-world, multi-component environments. And in biology, cell-level analyses failed to explain organism-level phenomena such as developmental or ecological interactions. Stated differently, in addition to reductionism to understand the parts, integration was needed to learn how the many parts fit into a functioning, organized whole.

It is easy to say integration is needed, but it is another thing altogether to do it practically. Though integration can be accomplished in many ways (e.g., equifinality), many AI algorithms and models were developed precisely to handle such multi-component complexity. In physics, neural-networks model turbulent flow and solve high-dimensional quantum systems. In chemistry, deep-learning models integrate reaction pathways to predict synthesis routes and molecular properties. And in biology, large-scale AI models reconcile genomic, proteomic, and phenotypic data to predict protein folding and regulatory interactions.

Behavior analysis is following a similar, albeit delayed, path. The end of the 19th and majority of 20th century saw behavior analytic researchers follow the reductionist pathway, doing their best to isolate respondent and operant phenomena in laboratory settings. The results have been textbooks filled with

behavioral principles and processes; journals filled with preparations to isolate particular phenomena; and practical techniques to deploy isolated behavioral principles developed under closed conditions to the messy, complex, everyday applied settings that are decidedly open systems.

The step behavior analysts are beginning to take is toward integration and complexity. How exactly do all these principles and processes fit back together in a behaving organism? What happens when we give dynamic action to the principles studied historically in isolated, steady-state preparations? How can we account for molecular and molar patterns of behavior simultaneously and over large swaths of time, as opposed to collecting data on and analyzing only minute-long or hour-long "sessions"? How do the contingencies contacted during one portion of a day (or week or month or year) interact with contingencies contacted during other portions of a day (or week or month or year)? How exactly does all this multivariable and multivariate control of organismic behavior integrate? As with other sciences, AI enables behavior scientists to more easily ask and answer these kinds of questions—assuming we can get our hands on the requisite data.

Changing *How* We Study

AI is also changing how scientists study their phenomena of interest. The prototypical research workflow likely involves some combination of: literature review, identification of a gap in our knowledge or understanding, designing the experimental methods to sufficiently ask the question(s) of interest, prepare the materials and calibrate the measurement instruments (and do our best to control for potential confounds), conduct the experiment, analyze the collected data, write up the results and how they fit within the larger literature, and disseminate the work.

AI is changing how this is done. In astronomy, machine-learning models scan massive sky-survey datasets to flag exoplanet candidates or transient events, leaving humans to interpret anomalies. In cellular and systems neuroscience, deep-learning tools track thousands of neurons, detect spikes, and run closed-loop experiments, while humans set goals and supervise what was once split across collection and analysis. In the social and behavioral sciences, LLMs map literatures, code open-ended responses, and draft preregistrations

or methods sections, enabling humans to focus on editing and conceptually refining the experimental approach.

Although this may seem like a simple efficiency upgrade, it is fundamentally different. "Deep research" tools can now turn a typed question into a document-length literature summary, bypassing the formerly visible steps of defining search strategies, screening articles, chaining references, and documenting inclusion decisions. The trade-off is opacity: the search parameters used by the LLM are hidden, inaccuracies go unchecked, and there is no record of what was missed. An auditable, replicable scholarly process risks becoming a few keystrokes entered into an inscrutable system. Researchers shift their behavior from methodical evidence-gathering toward scholarship that depends heavily on trusting AI-generated outputs.

Another way that AI is changing *how* scientists study is in the rise of computational and simulation-based research approaches. In physics, researchers use large-scale simulations for turbulence modeling and cosmological structure formation. In chemistry, molecular dynamics and reaction-pathway simulations predict folding, binding, and synthesis outcomes. And, in biology, agent-based and whole-cell simulations model development, evolution, and population dynamics. Though conducting this kind of research does not restrict one to using only AI, as we move toward the complex integration of an increasing set of independent and dependent variables, AI-specific techniques become increasingly useful.

How is this already impacting behavior analysis? Computational and simulation-based research is on the rise and seems likely to continue. Jack McDowell and his collaborators eagerly develop the *Evolutionary Theory of Behavior Dynamics*, and a similar "accounting for" of basic lab phenomena is underway using Peter Killeen's Mathematical Principles of Reinforcement. However, computational simulations aside, it's unclear whether researchers have adopted any tools that automatically conduct steps for them in the empirical chain (though it's also unclear whether this is a true absence of their use or merely a lack of authors reporting their use). But it seems easy to imagine a future where reduced response effort from using these tools increases their adoption in behavior analysis.

Changing Education and Training

You would be hard-pressed to find a teacher or educator today whose traditional lesson plans and teaching methods have not been upended by LLMs. For example, Forbes reported that 90% of college students use AI and rang the alarm bells that higher education needs AI fluency support now. Compounding this is the awkward fact that most of these students will be asked or required to use LLMs to complete their work post-graduation. The resulting variations in the stances one can take toward AI use in the classroom, alongside long-standing evidence that the way higher education conducts itself today needs to change, are causing an identity crisis among universities, colleges, and programs over the role of their academic programs in students' lives.

At least two camps within a larger, multidimensional spectrum probably exist relative to AI in higher education. One camp might view the function of education as equipping students with the entry-level skills needed to get and perform well in their first job after schooling. Here, AI would likely be embedded in every aspect of the curriculum. For example, students no longer need to devise a behavior assessment from scratch. Rather, they need to know how to prompt an LLM to generate a draft they can quickly tailor to their client. As another example, researchers may no longer need to know how to graph and visually analyze data. Rather, they need to know how to translate an intervention effect size from an AI system to make their decision. Under this purview, everything currently being taught to students would need to be reexamined for how the ongoing development of AI tools changes the nature of post-education employment.

A second camp might view the function of education differently. Here, the function of education is to provide foundational knowledge of fundamental concepts and principles, and to teach generalizable critical thinking skills. In turn, students will, hopefully, behave effectively in future work environments, no matter how those environments change over time, as these skills are fundamental. Under this purview, the data are fairly clear that domain novices who use AI while learning new material do not learn new domains well, even when they think they do. Thus, the pedagogical tactic here is to disallow the

use of AI altogether until students have demonstrated sufficient expertise in a topic (though what counts as 'sufficient expertise' is still unknown).

Though the second camp might seem like less work, it is not. Educators would still have to rework their courses to make them "AI-proof." This is a non-trivial task with few good models. Most classic academic tasks to assess knowledge gained are those that LLMs are fantastic at, such as summarizing an article or chapter, answering study guide questions, multiple choice quizzes, open-ended exam questions, creating and scripting presentations, or responding in discussion forums. Thus, students' temptation to use LLMs when studying and, if delivering classes online or asynchronously, to do some of this work directly, will be hard to resist and nearly impossible for educators to monitor without reverting to pen-and-paper work. Thus, even educators who disallow AI in their classes need to address contingencies that strongly favor LLM use.

In addition to the pedagogical considerations above, I want to return to the opening paragraphs of this essay as they relate to education and teaching. The analysis of data surrounding human behavior is ubiquitous in all modern industries because human behavior drives consumption of the product or service, all companies have employees who behave, or behavior change is the service provided by the company directly. Behavior analysts who are additionally skilled in technology, algorithms, and AI should be highly employable across all industries. By dramatically rethinking *what* and *how* we teach in this new age of AI, we might rapidly increase the presence of behavior analysts beyond the big three industries we filter into today.

Coda

Significant advances in technology can sometimes lead to punctuated changes in how scientists do their work, what they study, and how they teach future generations who will join them as scientists or carry that science into their work as practitioners within an industry. AI is—undoubtedly—augmenting, changing, enhancing, growing, improving, and upending nearly all sciences and their related industries. Current and future students will need unique skills to succeed in an AI-infused future. When shopping for careers during those critical educational years, students will likely want to choose a path that

will give them cutting-edge skills, bolstered by foundational knowledge, and the opportunity to succeed in the cultures and societies within which they will work and live. AI is currently applying selective pressures to all professions and scientific disciplines. And, as Skinner stated in *Contingencies of Reinforcement* in 1969 (p. 40), "If your culture has not convinced you [to work on its behalf], so much the worse for your culture." AI is cordially inviting us to work to uplevel "Behavior Analysis" as it has historically been known. Will we accept the invitation?

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About This Version

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